

Project 3 Report: Knowledge Graph-based Recommender System

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Abstract

This report presents a hybrid knowledge graph-based recommender system for click-through-rate prediction and top-5 recommendation. The proposed method combines popularity priors, local temporal preference estimation, knowledge-graph content profiles, item-based collaborative filtering, and a neural matrix factorization model. The trainable component is optimized by a mixed objective consisting of binary cross-entropy and Bayesian personalized ranking loss, so that both pointwise classification accuracy and pairwise ranking quality are considered. For CTR prediction, the final system learns a lightweight logistic fusion layer on an internal holdout split instead of relying on manually tuned user-ID-dependent weights. On the local user-aware 80/20 validation protocol, the final system achieves an AUC of 0.8570 and an nDCG@5 of 0.1456.

I Introduction

The objective of this project is to construct a recommender system with a scoring function

$$f(u, i) : \mathcal{U} \times \mathcal{I} \rightarrow \mathbb{R}, \quad (1)$$

where $u \in \mathcal{U}$ denotes a user and $i \in \mathcal{I}$ denotes an item. A larger score indicates a higher estimated preference of user u for item i . The system is evaluated by two complementary tasks. The CTR prediction task measures whether positive user-item interactions receive higher scores than negative interactions, using AUC. The top- k recommendation task requires returning a ranked list of $k = 5$ items for each user and is evaluated by nDCG@5.

In addition to interaction records, the project provides a knowledge graph (KG) about movie items. Each KG fact is represented as a triplet

$$(h, r, t) \in \mathcal{G}, \quad (2)$$

where h and t are entities and r is a typed relation such as genre, director, writer, language, country, rating, or star. Since item IDs in the interaction data correspond to entity IDs in the KG, the KG can be used as side information to reduce sparsity and improve item representation.

II Related Work

Classical recommender systems usually follow either collaborative filtering or content-based filtering. Collaborative filtering estimates user preference from interaction similarity, for example through matrix factorization or item-based nearest-neighbor models. Content-based recommendation instead describes items by observable attributes and recommends items whose attributes match a user's historical profile.

Knowledge graph-based recommendation extends content-based modeling by representing item attributes and semantic relations as structured triplets. In embedding-based KG recommenders, users, items, and

relations are mapped into latent vectors and trained by link prediction objectives. However, in a small CPU-only setting, a purely neural KG embedding model may be less stable and less time-efficient. Therefore, this project adopts a pragmatic hybrid architecture: sparse KG features and collaborative signals provide robust ranking priors, while matrix factorization supplies trainable personalized scores. A learned CTR fusion layer is used to calibrate heterogeneous scores for AUC, while the top-5 branch keeps a ranking-oriented score combination.

III Method

III-A Notation

Let $\mathcal{D}^+$ and $\mathcal{D}^-$ denote the observed positive and negative interaction sets:

$$\mathcal{D}^+ = \{(u, i, y) : y = 1\}, \quad \mathcal{D}^- = \{(u, i, y) : y = 0\}. \quad (3)$$

For each user u , let $\mathcal{P}_u$ and $\mathcal{N}_u$ be the sets of positively and negatively observed items. All feature scores are standardized by

$$z(x_i) = \frac{x_i - \mu_x}{\sigma_x + \epsilon}, \quad (4)$$

where μ_x and σ_x are computed over the candidate item scores.

III-B Popularity and Local Temporal Prior

The item-level popularity prior is computed by the log-count difference between positive and negative interactions:

$$s_{\text{pop}}(i) = \log(1 + n_i^+) - \log(1 + n_i^-), \quad (5)$$

where n_i^+ and n_i^- are the numbers of positive and negative training interactions of item i .

Because the released interactions are ordered by user ID, a local temporal prior is also used. For user u , define a window $\mathcal{W}(u)$ containing the most recent 128 training users before u . The local preference score is

$$s_{\text{tmp}}(u, i) = \log\left(1 + \sum_{v \in \mathcal{W}(u)} \mathbb{I}[(v, i) \in \mathcal{D}^+]\right) - \log\left(1 + \sum_{v \in \mathcal{W}(u)} \mathbb{I}[(v, i) \in \mathcal{D}^-]\right). \quad (6)$$

For a completely unseen user, this prior becomes the main cold-start ranking signal.

III-C Knowledge Graph Content Profile

Each item is converted into a sparse KG feature vector ϕ_i . A KG feature corresponds to a relation-neighbor pattern, for example (out, r, t) when item i appears as the head entity in (i, r, t) , and (in, r, h) when item i appears as the tail entity in (h, r, i) . The item feature vector is row-normalized:

$$\tilde{\phi}_i = \frac{\phi_i}{\|\phi_i\|_2}. \quad (7)$$

The user KG profile is constructed from positive and negative histories:

$$c_u = \frac{\sum_{i \in \mathcal{P}_u} \tilde{\phi}_i - \alpha \sum_{j \in \mathcal{N}_u} \tilde{\phi}_j}{\left\| \sum_{i \in \mathcal{P}_u} \tilde{\phi}_i - \alpha \sum_{j \in \mathcal{N}_u} \tilde{\phi}_j \right\|_2}, \quad \alpha = 0.75. \quad (8)$$

The KG content score is the cosine-style similarity

$$s_{\text{kg}}(u, i) = c_u^\top \tilde{\phi}_i. \quad (9)$$

III-D Item-based Collaborative Filtering

An item–item similarity matrix is estimated from positive user co-occurrence:

$$\text{sim}(i, j) = \frac{|\mathcal{U}_i^+ \cap \mathcal{U}_j^+|}{\sqrt{|\mathcal{U}_i^+|} \sqrt{|\mathcal{U}_j^+|}}, \quad (10)$$

where $\mathcal{U}_i^+$ is the set of users who positively interacted with item i . The collaborative filtering score is

$$s_{\text{cf}}(u, i) = \sum_{j \in \mathcal{P}_u} \text{sim}(i, j) - \beta \sum_{j \in \mathcal{N}_u} \text{sim}(i, j), \quad \beta = 0.5. \quad (11)$$

III-E Matrix Factorization with BCE and BPR

The trainable component is a matrix factorization model with user and item embeddings:

$$s_{\text{mf}}(u, i) = \mathbf{p}_u^\top \mathbf{q}_i + b_u + b_i + b_0, \quad (12)$$

where $\mathbf{p}_u, \mathbf{q}_i \in \mathbb{R}^{48}$ are latent vectors and b_u, b_i, b_0 are bias terms.

The pointwise binary cross-entropy loss is

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{|\mathcal{B}|} \sum_{(u, i, y) \in \mathcal{B}} [y \log \sigma(s_{\text{mf}}(u, i)) + (1 - y) \log(1 - \sigma(s_{\text{mf}}(u, i)))]. \quad (13)$$

To better optimize top- k ranking, Bayesian personalized ranking (BPR) is added. For each positive pair (u, i^+) , a negative item i^- is sampled from the user’s known negative set when available. The BPR loss is

$$\mathcal{L}_{\text{BPR}} = \frac{1}{|\mathcal{B}^+|} \sum_{(u, i^+, i^-)} \log(1 + \exp(s_{\text{mf}}(u, i^-) - s_{\text{mf}}(u, i^+))). \quad (14)$$

The final training objective is

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda \mathcal{L}_{\text{BPR}}, \quad \lambda = 0.7. \quad (15)$$

III-F Final Scoring Functions

For CTR prediction, the submitted system first extracts five candidate scores:

$$\mathbf{x}_{u, i} = [s_{\text{mf}}(u, i), s_{\text{tmp}}(u, i), s_{\text{kg}}(u, i), s_{\text{cf}}(u, i), s_{\text{pop}}(i)]. \quad (16)$$

Instead of using user-ID-dependent features such as the raw user index or a cold-user indicator, the final CTR branch learns a small logistic fusion model on an internal holdout split of the training data. Let $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ be the feature mean and standard deviation on this holdout set. The CTR score is

$$f_{\text{CTR}}(u, i) = b + \mathbf{w}^\top \frac{\mathbf{x}_{u, i} - \boldsymbol{\mu}}{\boldsymbol{\sigma} + \epsilon}. \quad (17)$$

The fusion weights $\mathbf{w}$ and bias b are optimized by binary cross-entropy with a small ℓ_2 penalty. If the learned fusion does not improve holdout AUC over the default robust mixture, the implementation falls back to the default mixture.

For top-5 recommendation, the score for an observed user is

$$f_{\text{top}}(u, i) = 0.50z(s_{\text{pop}}(i)) + 0.85z(s_{\text{tmp}}(u, i)) + 0.80z(s_{\text{kg}}(u, i)) + 0.70z(s_{\text{cf}}(u, i)) + 0.50z(s_{\text{mf}}(u, i)). \quad (18)$$

For a completely cold-start user, only the local temporal prior is used:

$$f_{\text{cold}}(u, i) = z(s_{\text{tmp}}(u, i)). \quad (19)$$

Algorithm 1 Hybrid KG-based Recommendation

Require: $\mathcal{D}^+$, $\mathcal{D}^-$, KG triplets $\mathcal{G}$

- 1: Build item popularity scores s_{pop}
 - 2: Build local temporal score table s_{tmp} with window size 128
 - 3: Extract sparse KG item features $\tilde{\phi}_i$ and user profiles c_u
 - 4: Build item similarity matrix $\text{sim}(i, j)$ from positive co-occurrence
 - 5: Train matrix factorization by minimizing $\mathcal{L}_{\text{BCE}} + 0.7\mathcal{L}_{\text{BPR}}$
 - 6: Learn CTR fusion weights on an internal holdout split
 - 7: **for** each CTR query (u, i) **do**
 - 8: Extract $\mathbf{x}_{u,i}$ and return $f_{\text{CTR}}(u, i)$
 - 9: **end for**
 - 10: **for** each top-5 query user u **do**
 - 11: Score all items by f_{top} or f_{cold}
 - 12: Remove items in $\mathcal{P}_u$
 - 13: Return the five items with largest scores
 - 14: **end for**
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Items appearing in the user’s training positive set are filtered out before selecting the top-5 list.

IV Experiments

IV-A Dataset

The dataset is derived from MovieLens-1M. The released interaction data contains 26,638 positive records and 24,037 negative records. User IDs range from 0 to 6014, and item IDs range from 0 to 2346. The KG contains 20,195 triplets, 6,729 entities, and seven relation types: star, genre, writer, director, rating, language, and country.

TABLE I: Dataset statistics

Quantity	Value
Positive interactions	26,638
Negative interactions	24,037
Users	6,015
Items	2,347
KG triplets	20,195
KG relation types	7

IV-B Evaluation Protocol

The original arrays are nearly sorted by user ID. A naive chronological split would place many high-ID users only in the test set, turning top-5 evaluation into a cold-start problem. Therefore, the local validation protocol uses a user-aware 80/20 split. For each user, one record is first reserved in the training set to guarantee user visibility. The remaining records are globally shuffled and split so that the final training ratio remains close to 80%.

Under this split, the test set contains 5,064 users and no cold-start test user. The top-5 task evaluates 3,649 users with positive test items, and all of them have appeared in the training set.

TABLE II: User-aware 80/20 split statistics

Subset	Train	Test	Train ratio
Positive interactions	21,310	5,328	0.79998
Negative interactions	19,229	4,808	0.79998

IV-C Metrics

AUC is used for CTR prediction:

$$\text{AUC} = \frac{1}{|S^+||S^-|} \sum_{x^+ \in S^+} \sum_{x^- \in S^-} \mathbb{I}[f(x^+) > f(x^-)]. \quad (20)$$

nDCG@5 is used for top-5 recommendation:

$$\text{DCG@5} = \sum_{j=1}^5 \frac{\mathbb{I}[i_j \in \mathcal{T}_u^+]}{\log_2(j+1)}, \quad (21)$$

$$\text{nDCG@5} = \frac{\text{DCG@5}}{\text{IDCG@5}}, \quad (22)$$

where $\mathcal{T}_u^+$ is the positive test item set of user u .

IV-D Results

TABLE III: Final local validation performance

Metric	Value	Grading threshold
AUC	0.8570	0.7000
nDCG@5	0.1456	0.1200

The final model exceeds the highest threshold for both AUC and nDCG@5. The measured runtime is also well below the project limits: initialization takes about 0.66 seconds, training including CTR fusion takes about 4.47 seconds, CTR evaluation takes about 1.22 seconds, and top-5 evaluation takes about 4.38 seconds.

TABLE IV: Effect of matrix factorization training epochs before learned CTR fusion

Epochs	AUC	nDCG@5	Training time (s)
4	0.7580	0.1415	0.36
8	0.7578	0.1420	0.65
12	0.7584	0.1431	1.00
16	0.7595	0.1435	1.36
20	0.7606	0.1456	1.69
24	0.7614	0.1451	2.07
32	0.7624	0.1447	2.60

Before adding the learned CTR fusion layer, increasing the number of epochs improves AUC steadily, but nDCG@5 peaks at 20 epochs and slightly decreases afterwards. This suggests that longer training improves pointwise discrimination but may start to overfit the ranking objective. Therefore, 20 epochs are used for matrix factorization, and the final CTR calibration is then learned on an internal holdout split.

IV-E External MovieLens-1M Sanity Check

To test whether the model behavior is reasonable beyond the released project arrays, an additional sanity check is conducted on the original MovieLens-1M files. Ratings greater than or equal to 4 are treated as positive implicit feedback. For the implicit setting, negative samples are drawn from movies that the corresponding user has not rated. Movie genre, year, and decade are converted into simple KG triplets. This external test is not part of the official grading protocol, but it helps verify that the model is not only exploiting the specific project split.

TABLE V: External MovieLens-1M validation

Negative sampling protocol	AUC	nDCG@5
Low explicit ratings as negatives	0.8917	0.2348
Unrated movies as negatives	0.9388	0.2798

The model obtains high AUC and nDCG@5 under both protocols. The unrated-negative protocol is closer to an implicit-feedback recommendation scenario and gives stronger performance, indicating that the hybrid signals generalize to a standard recommendation benchmark.

IV-F Comparison with a RotatE Variant

As an additional reference, a RotatE-style KG embedding variant was implemented and evaluated. In this variant, each user is treated as an extra entity with ID offset $|\mathcal{E}|$, each item remains its original KG entity, and user feedback is represented by an additional relation r_{fb} . The RotatE score is

$$s_{\text{RotatE}}(h, r, t) = \gamma - \|\mathbf{e}_h \circ \mathbf{r}_r - \mathbf{e}_t\|, \quad (23)$$

where $\mathbf{r}_r$ is a complex-valued rotation with $|\mathbf{r}_r| = 1$. The training objective combines BCE, BPR, and a lightweight KG link-prediction regularizer:

$$\mathcal{L}_{\text{RotatE}} = \mathcal{L}_{\text{BCE}} + 0.7\mathcal{L}_{\text{BPR}} + 0.05\mathcal{L}_{\text{KG}}. \quad (24)$$

TABLE VI: Comparison between the learned-fusion final model and the RotatE variant

Model	AUC	nDCG@5	Training time (s)	Top-5 time (s)
MF+BPR hybrid with learned CTR fusion	0.8570	0.1456	4.47	4.38
RotatE variant	0.7527	0.1401	13.45	33.16

The RotatE variant is theoretically more expressive for KG link prediction, but it is less effective in this recommendation setting. It requires scoring all candidate items through a complex-valued relation transformation during top-5 evaluation, which substantially increases CPU inference time. Moreover, the target task is user-item ranking rather than pure KG completion, so the simpler matrix factorization component trained with BCE+BPR provides a more direct inductive bias. Therefore, the final submitted system keeps the MF+BPR model, uses KG information through sparse content profiles, and calibrates the CTR branch with learned fusion weights.

V Discussion

The main advantage of the proposed system is that it combines heterogeneous evidence sources. Popularity and local temporal priors are robust for sparse or cold-start cases. KG content profiles exploit semantic item at-

tributes and make recommendations less dependent on exact co-occurrence. Item-based collaborative filtering captures neighborhood-level preference transfer. Matrix factorization adds a trainable latent representation, and the BPR term directly encourages positive items to be ranked above negative items.

The current implementation also has limitations. First, the CTR fusion layer is learned only as a shallow logistic model rather than as an end-to-end neural architecture. Second, the KG component is based on sparse relation-neighbor features instead of deep relational message passing. Third, the local temporal prior uses user ID order as an empirical signal, which is effective for the provided data but may not generalize if the test distribution is randomly permuted. To reduce this risk in CTR prediction, the final system avoids raw user-ID and cold-user indicator features and relies on learned fusion of interaction, KG, CF, temporal, and popularity scores.

VI Conclusion

This project implements a hybrid KG-based recommender system for CTR prediction and top-5 recommendation. The final algorithm integrates popularity, temporal locality, KG content profiles, item-based collaborative filtering, BCE+BPR-trained matrix factorization, and a learned logistic CTR fusion layer. A user-aware 80/20 validation protocol is adopted to avoid artificial cold-start bias caused by sorted interaction arrays. The final system obtains an AUC of 0.8570 and an nDCG@5 of 0.1456, satisfying the highest score thresholds under the local validation setting while remaining computationally efficient on CPU.